Project Plan

Imperial College London
Department of Earth Science and Engineering
MSc Environmental Data Science and Machine Learning

Comparison of Direct and Cascaded High-Resolution Spatial Downscaling of Wildfire Data

Author:
Tsuyu Huang
tsuyu.huang25@imperial.ac.uk

Supervisors:
Jorge Veiras (jorge@artiocarbon.com)
Prof. Anna Korre (a.korre@imperial.ac.uk)

Company:
Artio
www.artiocarbon.com

June 7, 2026

Abstract

Wildfires are destructive and a rapidly evolving hazard. With climate change, they become increasingly severe and frequent. This increases the need for a higher-resolution wildfire data for better monitoring and risk assessment. Geostationary satellites (GEO) have high-temporal resolution, but lacks the resolution needed for a detailed wildfire risk assessment. While low orbit satellites (LEO) provide better spatial resolution, but low temporal resolution to monitor rapid wildfire spread. Existing generative downscaling models have demonstrated their effectiveness in producing sharp, high-resolution remote sensing data. But many still suffers from artefacts, blurriness, and hallucinations. This project aims to achieve a high-resolution spatial downscaling of wildfire data using downscaling models that integrates additional physical covariates, such as topography and landcover to mitigate physically implausible outputs. The project compares direct (2km to 30m) and cascaded (2km to 375m to 30m) downscaling pipelines and their effectiveness in producing high-fidelity, high-resolution wildfire data. The project compares diffusion (EDiffSR), attention U-Net (HAUNet), and transformer-based (ESTNet) downscaling models.

Contents

1 Problem Description and Motivation **2**

2 Literature Review **2**

2.1 Current State of Remote Sensing Downscaling 2

2.2 Selected Downscaling Architectures for Comparison 3

3 Methodology **3**

4 Progress to Date **3**

5 Future Plans **3**

1 Problem Description and Motivation

In recent years, due to rising popularity in Net-Zero initiatives, carbon investors are increasingly funding carbon reduction and removal projects, such as reforestation (Lambert and Turner, 2024). However, these carbon projects are at risk because wildfires can destroy and release the carbon back into the atmosphere. With climate change, wildfires are becoming increasingly severe and frequent (Jones et al., 2020). Which increases the need for better wildfire monitoring and prediction for risk assessment. This means high-spatial and high-temporal resolution wildfire data is highly desired. However, high-temporal resolution data obtained from geostationary satellites (GEO) have limited spatial resolution, which makes them difficult to use for accurate risk assessment. While low orbit satellites (LEO) offers high-spatial resolution, but low-temporal resolution.

To bridge this gap, existing methods uses Deep Learning (DL) models for downscaling due to their ability in recognising non-linear spatiotemporal patterns. However, the resolution gap between GEO and LEO is significant (2km - 30m). This increases the risk of producing physically implausible or unrealistic outputs, as a single low-resolution (LR) pixel can correspond to different high-resolution (HR) details (Z. Wang et al., 2020). In this project, to mitigate this limitation, the model integrates additional physical information, such as HR topography and landcover to help guide the downscaling process. To further improve downscaling, the project aims to investigate whether cascaded downscaling (2km to 375m to 30m) produces more accurate and better results compared to direct downscaling (2km to 30m). By breaking up the downscaling process, the model trains on a smaller resolution gap, aiming to mitigate physically implausible outputs. This project will compare the cascaded and direct pipeline performance of different downscaling architectures, including hybrid attention U-Net, a transformer-based model, and a diffusion-based model.

2 Literature Review

2.1 Current State of Remote Sensing Downscaling

Traditional downscaling methods often fail to capture the complex non-linear patterns and variability in remote sensing data. In recent years, DL methods have shown promising results that outpaces traditional approaches (Sdraka et al., 2022). In particular, Convolutional Neural Networks (CNNs) and U-Net architectures have been widely used for their ability to capture spatial features. Generative Adversarial Networks (GANs) have also been used for their ability to generate sharp images. Recently, Transformer and Diffusion-based approaches have been proposed in super-resolution research. Transformers use attention mechanisms to capture long-range dependencies (J. Wang et al., 2025), while Diffusion models shows good generative ability and fidelity (Y. Wang et al., 2023).

However, many of these approaches have limitations, particularly in preserving fine-grained textures and geometric structures (Qi et al., 2026). This is problematic for wildfire data, where the spatial patterns and textures are crucial for accurate risk assessment. To address these limitations, there is a need for domain-specific architectures that can better capture the characteristics of wildfire data.

In the current literature, there is also a lack of studies that compares the performance of direct and cascaded downscaling pipelines. Cascaded downscaling has the potential to mitigate the issue of physically implausible outputs by breaking up the downscaling process into smaller steps. Some studies have explored cascaded approach, such as Jing et al., 2022, and Chen et al., 2023, but more research is needed to fully understand its benefits and limitations in the context of wildfires. However, it also introduces additional complexity and computational cost. Therefore, research is needed to investigate whether there are significant benefits to cascaded downscaling.

2.2 Selected Downscaling Architectures for Comparison

Among CNN-based approaches, HAUNet (J. Wang et al., 2023) uses a hybrid attention mechanism to fuse multiscale, local and global features. This allows it to capture both fine-grained details and broader contextual information. HAUNet will be selected as one of the architectures for comparison. Because while U-Nets capture spatial features effectively, combining them with attention mechanism makes it a strong baseline for capturing spatially heterogeneous patterns in wildfire data.

ESTNet (Kang et al., 2024) is a transformer-based architecture that uses efficient spatial-temporal attention to capture long-range dependencies in the data. It is based on Swin-Transformers with group attention. It uses multi-window self-attention for better efficiency without sacrificing reconstruction quality. For transformer-based approach, ESTNet will be selected for comparison due to their ability in capturing long-range dependencies, making them a suitable candidate for downscaling wildfire data, which has complex spatial patterns. ESTNet specifically is preferred over alternatives such as SwinIR (Liang et al., 2021) because their efficiency makes them more suitable for the large amount of data and inputs used in this project.

EDiffSR (Xiao et al., 2023) is a conditional diffusion-based model. It uses simplified channel attention for better noise prediction quality, and Conditional Prior Enhancement Module (CPEM) to extract the most valuable information from the LR input. EDiffSR is selected as the diffusion-based model for comparison because it effectively leverages prior information and promotes the condition with more informative input.

3 Methodology

The selected architectures (HAUNet, ESTNet, EDiffSR) for comparison is tentatively selected based on their performance in super-resolution tasks, and their suitability for the specific requirements of the wildfire downscaling. However, the final selection may be adjusted as the project progresses.

4 Progress to Date

5 Future Plans

References

Chen, H.-W., Xu, Y.-S., Hong, M.-F., Tsai, Y.-M., Kuo, H.-K., & Lee, C.-Y. (2023). Cascaded Local Implicit Transformer for Arbitrary-Scale Super-Resolution, 18257–18267. Retrieved June 7, 2026, from https://openaccess.thecvf.com/content/CVPR2023/html/Chen_Cascaded_Local_Implicit_Transformer_for_Arbitrary-Scale_Super-Resolution_CVPR_2023_paper.html

Jing, Y., Lin, L., Li, X., Li, T., & Shen, H. (2022). Cascaded Downscaling–Calibration Networks for Satellite Precipitation Estimation. *IEEE Geoscience and Remote Sensing Letters*, *19*, 1–5. https://doi.org/10.1109/LGRS.2022.3214083

Jones, M. W., Smith, A. J. P., Betts, R., Canadell, J. G., Prentice, I. C., & Quéré, C. L. (2020). Climate Change Increases the Risk of Wildfires.

Kang, X., Duan, P., Li, J., & Li, S. (2024). Efficient Swin Transformer for Remote Sensing Image Super-Resolution. *IEEE Transactions on Image Processing*, *33*, 6367–6379. https://doi.org/10.1109/TIP.2024.3489228

Lambert, J., & Turner, G. (2024, November 14). *Investment Trends and Outcomes in the Global Carbon-Credit Market-2024 | MSCI*. Retrieved June 5, 2026, from https://www.msci.com/research-and-insights/paper/investment-trends-and-outcomes-in-the-global-carbon-credit-market-2024

Liang, J., Cao, J., Sun, G., Zhang, K., Gool, L. V., & Timofte, R. (2021, August 23). *SwinIR: Image Restoration Using Swin Transformer*. arXiv: 2108.10257 [eess.IV]. https://doi.org/10.48550/arXiv.2108.10257

Qi, Y., Lou, M., Liu, Y., Li, L., Yang, Z., & Nie, W. (2026). Advancing image super-resolution techniques in remote sensing: A comprehensive survey. *ISPRS Journal of Photogrammetry and Remote Sensing*, *231*, 68–100. https://doi.org/10.1016/j.isprsjprs.2025.10.024

Sdraka, M., Papoutsis, I., Psomas, B., Vlachos, K., Ioannidis, K., Karantzalos, K., Gialampoukidis, I., & Vrochidis, S. (2022). Deep Learning for Downscaling Remote Sensing Images: Fusion and super-resolution. *IEEE Geoscience and Remote Sensing Magazine*, *10*(3), 202–255. https://doi.org/10.1109/MGRS.2022.3171836

Wang, J., Chen, M., Karaev, N., Vedaldi, A., Rupprecht, C., & Novotny, D. (2025, March 14). *VGGT: Visual Geometry Grounded Transformer*. arXiv: 2503.11651 [cs.CV]. https://doi.org/10.48550/arXiv.2503.11651

Wang, J., Wang, B., Wang, X., Zhao, Y., & Long, T. (2023). Hybrid Attention-Based U-Shaped Network for Remote Sensing Image Super-Resolution. *IEEE Transactions on Geoscience and Remote Sensing*, *61*, 1–15. https://doi.org/10.1109/TGRS.2023.3283769

Wang, Y., Yang, W., Chen, X., Wang, Y., Guo, L., Chau, L.-P., Liu, Z., Qiao, Y., Kot, A. C., & Wen, B. (2023, November 23). *SinSR: Diffusion-Based Image Super-Resolution in a Single Step*. arXiv: 2311.14760 [cs.CV]. https://doi.org/10.48550/arXiv.2311.14760

Wang, Z., Chen, J., & Hoi, S. C. H. (2020, February 8). *Deep Learning for Image Super-resolution: A Survey*. arXiv: 1902.06068 [cs.CV]. https://doi.org/10.48550/arXiv.1902.06068

Xiao, Y., Yuan, Q., Jiang, K., He, J., Jin, X., & Zhang, L. (2023, October 30). *EDiffSR: An Efficient Diffusion Probabilistic Model for Remote Sensing Image Super-Resolution*. arXiv: 2310.19288 [eess.IV]. https://doi.org/10.48550/arXiv.2310.19288